BOYONG ZHAO

[+8618223511409] | zby_18223511409@163.com

EDUCATION

Harbin Institute of Technology, Shenzhen Shenzhen, China
Bachelor of Engineering in Automation Sep. 2021 - Jul. 2025

- **GPA:** 89/100 (3.745/4.000), **Rank:** 26/255 (10.2%)
- **Language:** IELTS 6.0
- **Research Interests:** My research interest lies in the intersection of robotics, perception and SLAM, with their application in complex, real-world environments.

ACADEMIC PAPER

- **Boyong Zhao**, Yuan Shen, Pui Yi Chui, Yuzhe Li, Junfeng Wu (2026). "AquaPoint: Self-Supervised Cross-Domain Framework for Robust Underwater Local Features." *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, **accepted**.

RESEARCH EXPERIENCE

- **Research on Robust Underwater Local Features** Jun. 2025 - May 2026
  LIAS The Chinese University of Hong Kong, Shenzhen
  Advisor: Prof. Junfeng Wu School of Data Science
  - **Introduction:**Proposed AquaPoint, a self-supervised cross-domain framework integrating physics-guided domain bridging and geometry-consistent descriptor alignment to tackle severe feature tracking failures caused by underwater visual degradation.
  - **Key Contributions:**
    * Designed the Physics-Guided Domain Bridging (PGDB) module, utilizing physics-consistent hydro-optical degradation synthesis and OceanSim simulations to explicitly mitigate photometric and structural domain gaps.
    * Engineered the Geometry-Consistent Descriptor Alignment (GCDA) module, leveraging RAFT optical flow and multi-view geometric verification (temporal, stereo, and epipolar checks) to mine reliable pseudo-correspondences for contrastive learning.
    * Integrated the AquaPoint front-end into a decoupled pure-odometry ORB-SLAM3 configuration using LightGlue matching for downstream evaluation.
  - **Outcome:** Wrote the paper *AquaPoint: Self-Supervised Cross-Domain Framework for Robust Underwater Local Features* (as the first author).

- **Research on Multimodel Model Perception under Challenging Scenarios** Sep. 2024 - Jan. 2025
  ClassLab Harbin Institute of Technology, Shenzhen
  Advisor: Prof. Liang Hu School of Intelligence Science and Engineering
  Participated in the optimization of long-tail autonomous driving scenarios by fine-tuning the LLaVA vision-language large model. Explored transfer learning strategies to enhance complex multimodal perception and decision-making.

- **Research on Reinforcement Learning** July. 2023 - Sep. 2023
  RLGroup Harbin Institute of Technology, Shenzhen
  Advisor: Yanjie Li School of Intelligence Science and Engineering
  Investigated and implemented classical reinforcement learning algorithms (e.g., Q-learning, DQN) within simulation environments.

HONORS AND AWARDS

- **Honors:**
  - **Elite Research Assistant of School of Data Science, CUHKSZ** 2026
  - **Second Class Academic Scholarship** 2022
  - **Second Class Academic Scholarship** 2023
  - **Third Class Academic Scholarship** 2024
  - **Outstanding Student** 2023
- **Competition Awards:**
  - **Thrid Prize in National College Mathematical Modeling Competition** 2022.9

SKILLS

Programming and Tools: C/C++, Python, Matlab, LaTeX, Linux, ROS